TIME SERIES ANALYSIS

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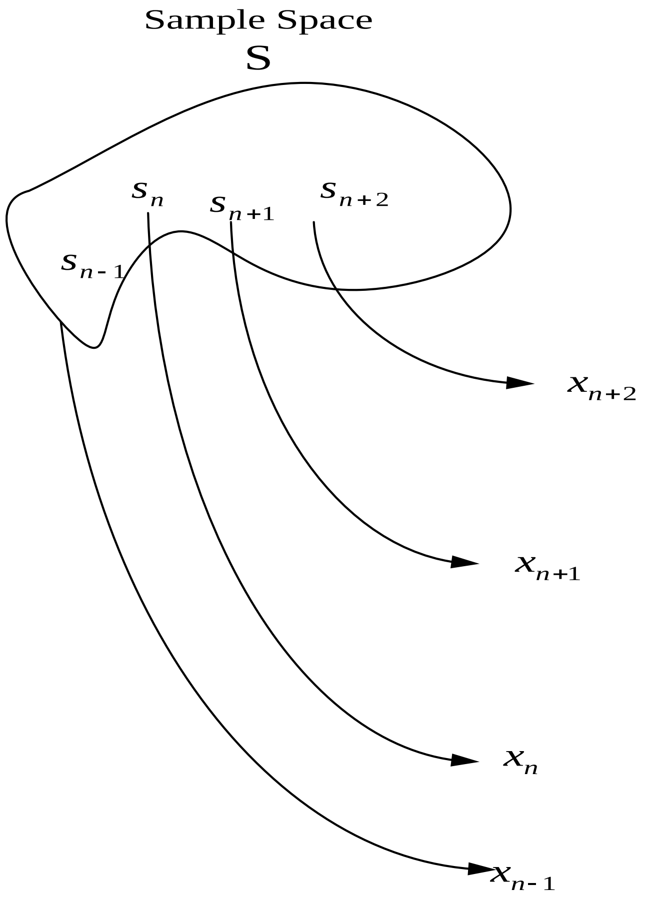
Random Processes

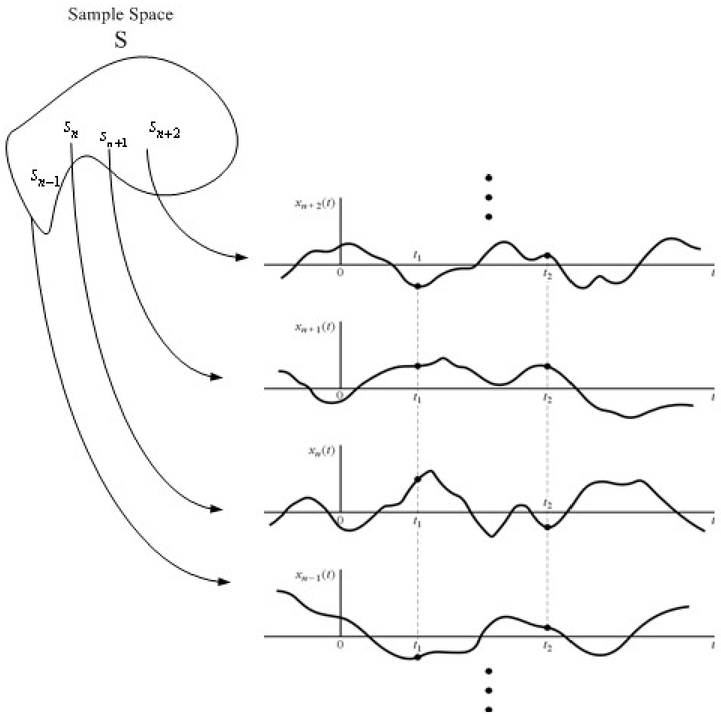
- A **RANDOM VARIABLE** X, is a rule for assigning to every outcome, ω, of an experiment a number $X(\omega)$.
  - Note: X denotes a random variable and $X(\omega)$ denotes a particular value.
- A **RANDOM PROCESS** $X(t)$ is a rule for assigning to every ω, a function $X(t,\omega)$.
  - Note: for notational simplicity we often omit the dependence on ω.

Random Process

The concept of random variable was defined previously as mapping from the **Sample Space S** to the real line as shown below



The concept of random process can be extended to include time and the outcome will be random functions of time as shown below



The functions $\cdots x_{n+2}(t), x_{n+1}(t), x_n(t), x_{n-1}(t), \cdots$ **are one realizations**

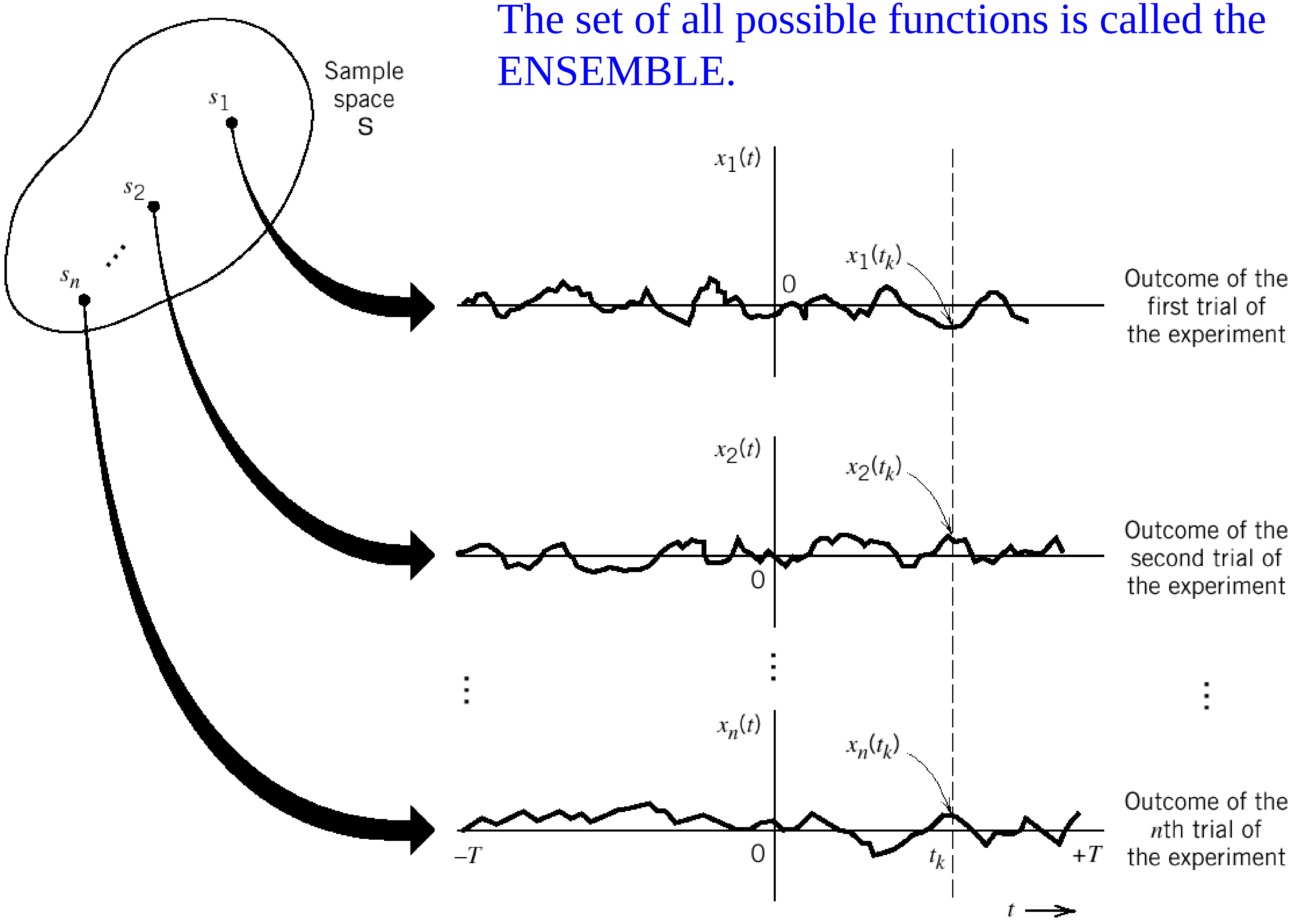
of many of the random process X(t)

A random process also represents a random variable when time is fixed

$X(t_1)$ is a random variable

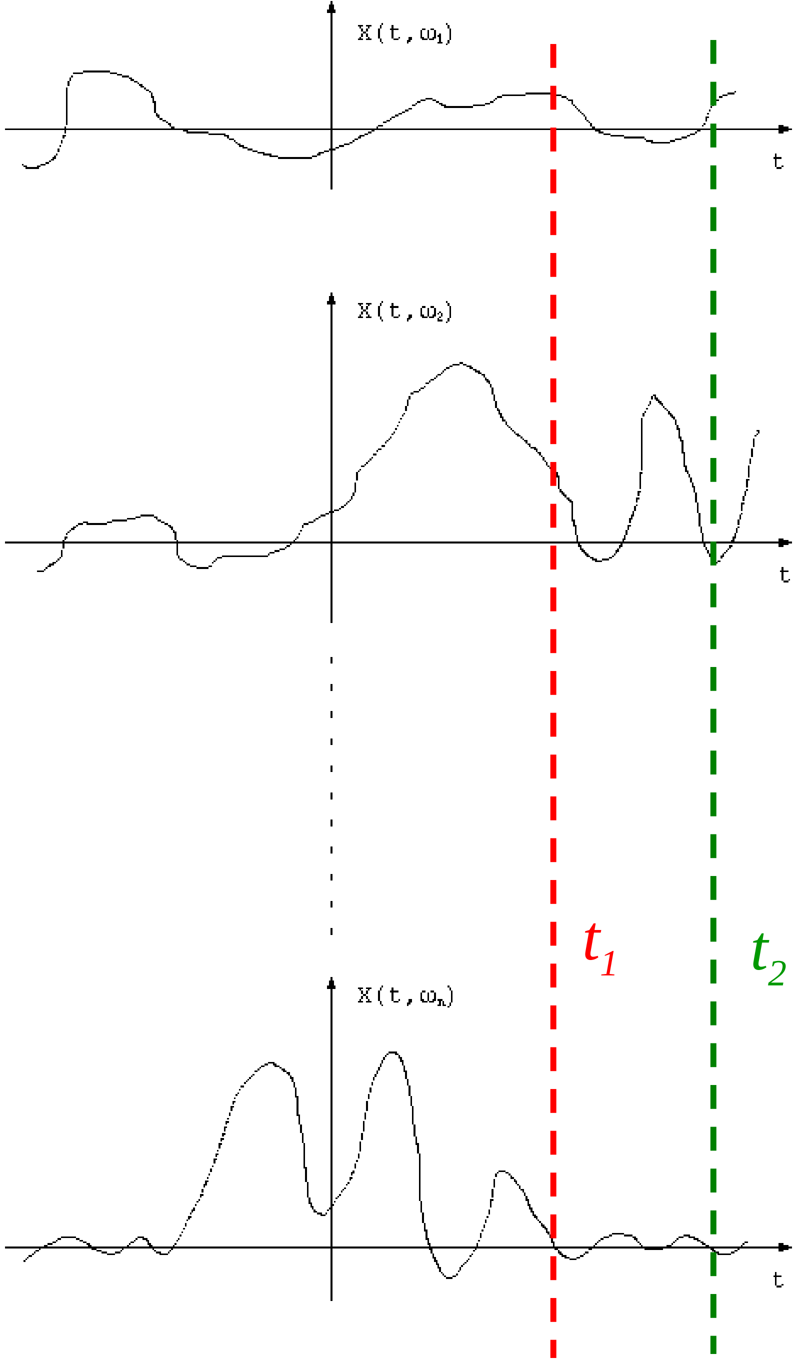
Ensemble of Sample Functions

The set of all possible functions is called the ENSEMBLE.



Random Processes

- A general Random or Stochastic Process can be described as:
  - Collection of time functions (signals) corresponding to various outcomes of random experiments.
  - Collection of random variables observed at different times.
- Examples of random processes in communications:
  - Channel noise,
  - Information generated by a source,
  - Interference.



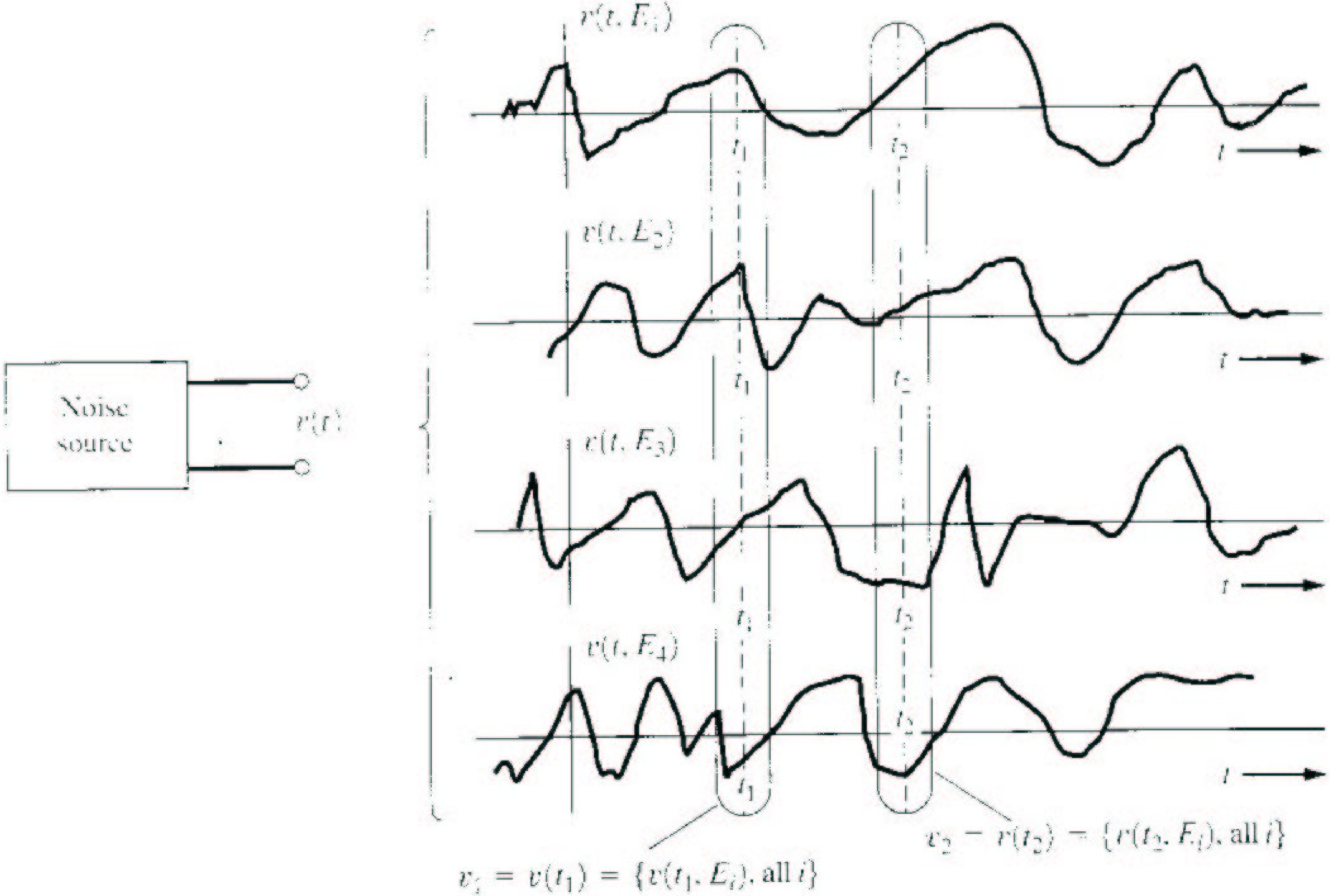


Figure 6–1 Random-noise source and some sample functions of the random-noise process.

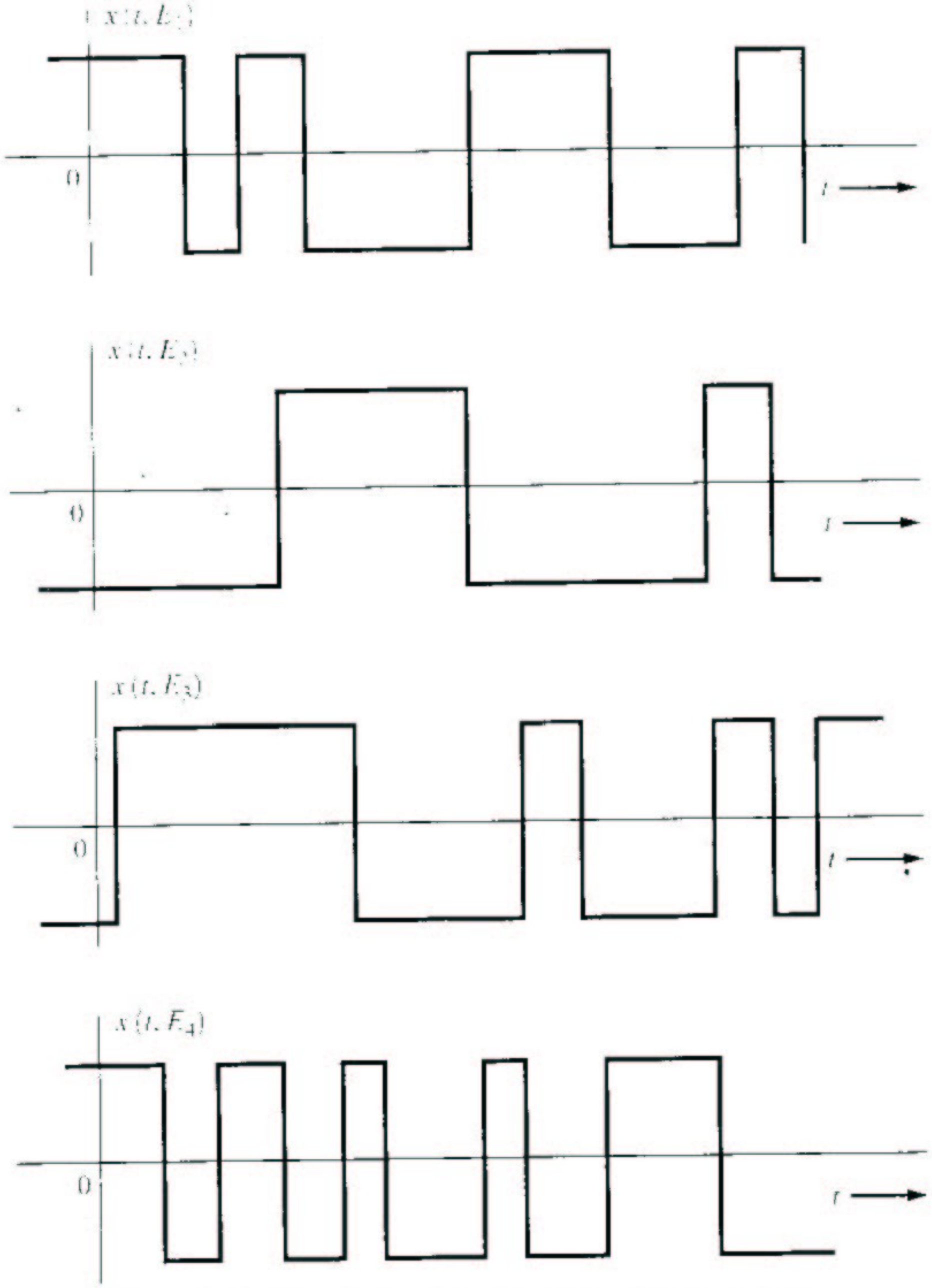


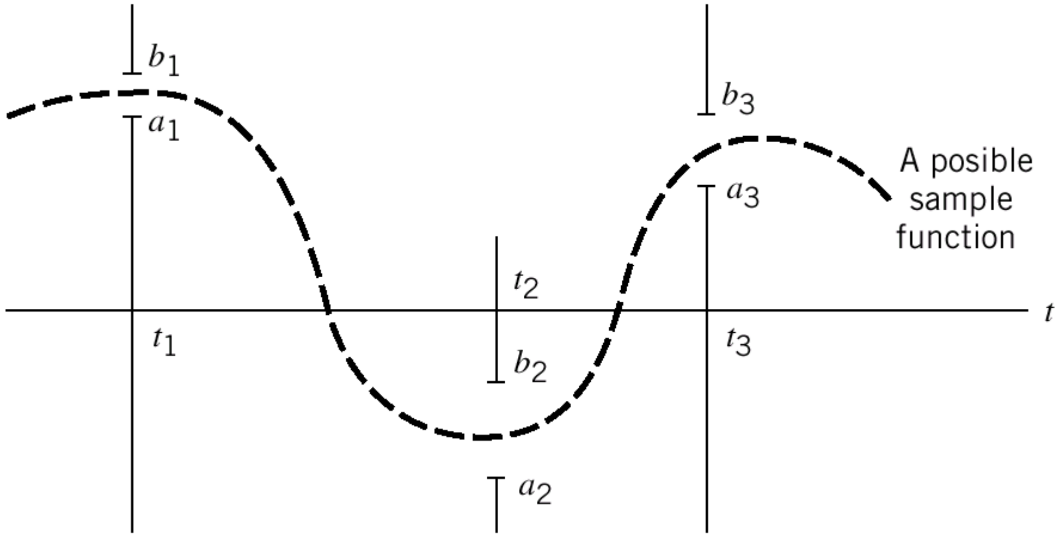
Figure 6–2 Sample functions of a binary random process.

Collection of Time Functions

- Consider the time-varying function representing a random process where ω_i represents an outcome of a random event.
- Example:
  - a box has infinitely many resistors ($i=1,2,\ldots$) of same resistance R.
  - Let ω_i be event that the ith resistor has been picked up from the box
  - Let $v(t, \omega_i)$ represent the voltage of the thermal noise measured on this resistor.

Collection of Random Variables

- For a particular time $t=t_o$ the value $x(t_o,\omega_i)$ is a random variable.
- To describe a random process we can use collection of random variables $\{x(t_o,\omega_1)\ ,\ x(t_o,\omega_2)\ ,\ x(t_o,\omega_3)\ , \ldots \}$.
- Type: a random processes can be either discrete-time or continuous-time.
- Probability of obtaining a sample function of a RP that passes through the following set of windows. Probability of a joint event.



Description of Random Processes

- **Analytical description:** $X(t) = f(t, \omega)$ where ω is an outcome of a random event.
- **Statistical description:** For any integer N and any choice of $(t_1, t_2, \ldots, t_N)$ the joint pdf of $\{X(t_1), X(t_2), \ldots, X(t_N)\}$ is known. To describe the random process completely the PDF $f(\mathbf{x})$ is required.

$$x_1 = x(t_1), \quad \mathbf{x} = [x_1, x_2, \ldots x_N]$$

$$f(\mathbf{x}) = f\{x(t_1), x(t_2), \ldots.\ x(t_N)\}$$

Example: Analytical Description

- Let $X(t) = A\cos(2\pi f_0 t + \theta)$s a random variable uniformly distributed on $[0,2\pi]$.
- Complete statistical description of $X(t_o)$ is:
  - Introduce $Y = 2\pi f_0 t_0 + \theta$
  - Then, we need to transform from y to x:

$$p_X(x)dx = p_Y(y_1)dy + p_Y(y_2)dy$$

  - We need both y_1 and y_2 because for a given x the equation $x=A\cos(y)$ has two solutions in $[0,2\pi]$.

Analytical (continued)

- Note x and y are actual values of the random variables X and Y.
- Since $\left|\frac{dx}{dy}\right| = |A \sin y| = \left|\sqrt{A^2 - x^2}\right|$

  and p_Y is uniform in $[2\pi f_0 t_0, 2\pi f_0 t_0 + 2\pi]$, we get

$$p_X(x) = \begin{cases} \dfrac{1}{\pi\sqrt{A^2 - x^2}} & -A < x < A \\ 0 & \text{Elsewhere} \end{cases}$$

- Using the analytical description of $X(t)$, we obtained its statistical description at any time t.

Example: Statistical Description

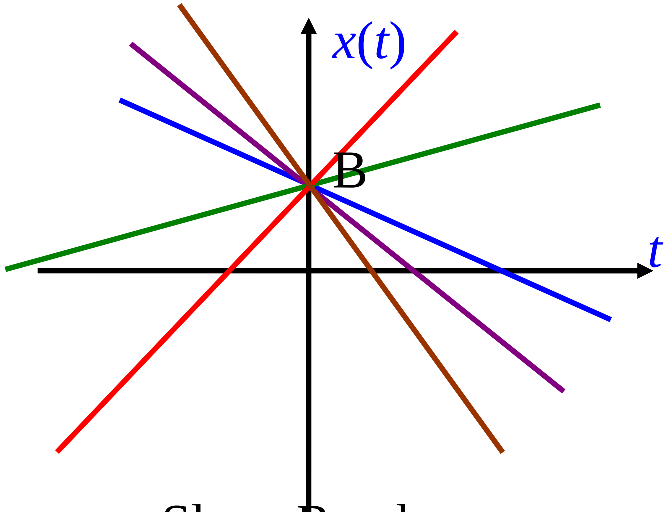
- Suppose a random process x(t) has the property that for any N and $(t_0, t_1, \ldots, t_N)$ the joint density function of $\{x(t_i)\}$ is a jointly distributed Gaussian vector with zero mean and covariance

$$\sigma_{ij} = \sigma^2 \min(t_i, t_j)$$

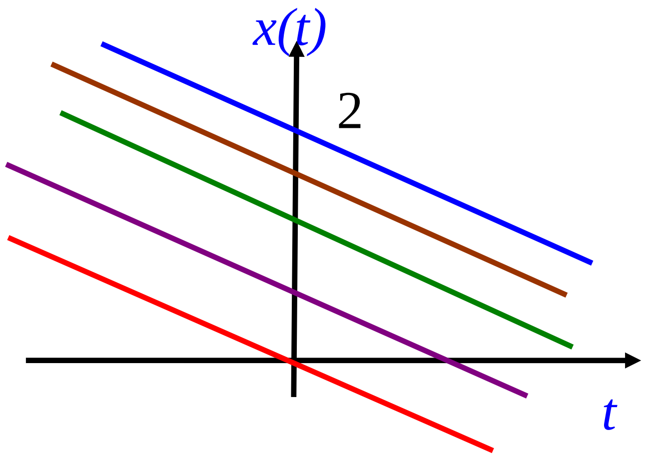
- This gives complete statistical description of the random process x(t).

Activity: Ensembles

- Consider the random process: x(t)=At+B
- Draw ensembles of the waveforms:
  - B is constant, A is uniformly distributed between [-1,1]
  - A is constant, B is uniformly distributed between [0,2]
- Does having an "Ensemble" of waveforms give you a better picture of how the system performs?



Slope Random



B intersect is Random

Stationarity

- Definition: A random process is STATIONARY to the order N if for any $t_1, t_2, \ldots, t_N$,

$$f_x\{x(t_1), x(t_2), \ldots x(t_N)\} = f_x\{x(t_1+t_0), x(t_2+t_0), \ldots, x(t_N+t_0)\}$$

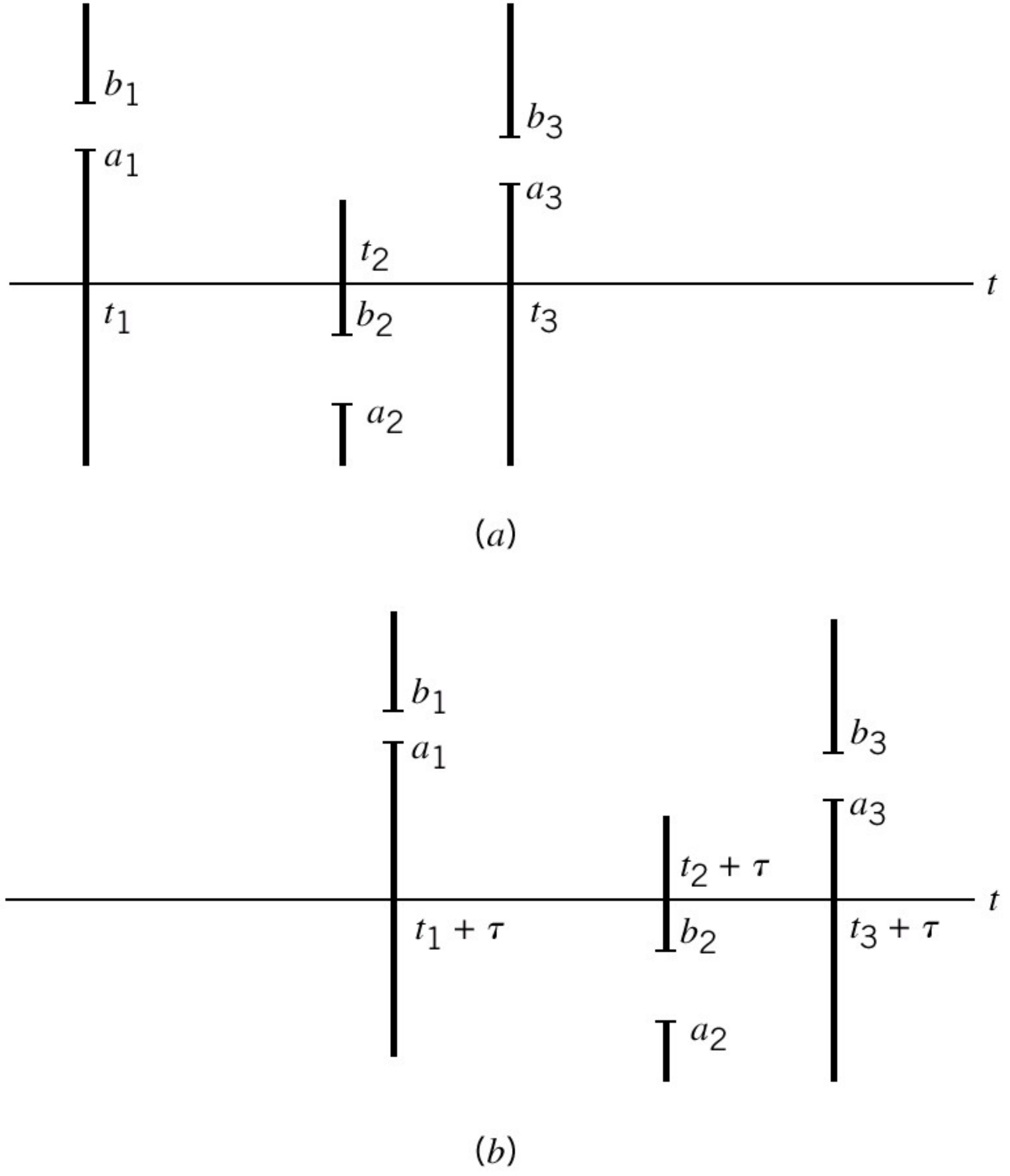
- This means that the process behaves similarly (follows the same PDF) regardless of when you measure it.

- A random process is said to be STRICTLY STATIONARY if it is stationary to the order of $N \rightarrow \infty$.

- Is the random process from the coin tossing experiment stationary?

Illustration of Stationarity

Time functions pass through the corresponding windows at different times with the same probability.



Properties of stochastic process

Stationarity

Consider a discrete time-series $\{x(t_1), x(t_2), \cdots, x(t_N)\}$
The joint probability distribution function (pdf) of $\{x(t_k)\}_{k=1}^{N}$ describes the probability that the random variable X takes on the values $X = x(t_k)$ jointly in a sequence of samples.

A stochastic process is said to be strictly stationary if the joint pdf associated with the N observations taken at $t_1, t_2, \cdots, t_N$ is identical to the joint pdf associated with another set of N observations taken at times $t_1 + k, t_2 + k, \cdots, t_N + k$

Stationarity

Stationarity

A discrete-time is strictly stationary if the joint pdf of a set of observations remains unaffected by shifting all the times of observation forward or backward by an integer amount k.

Stationarity implies that properties of the pdf such as mean, variance, and higher-order moments do not change with time.

STATIONARITY

- The most vital and common assumption in time series analysis.
- The basic idea of stationarity is that the probability laws governing the process **do not** change with time.
- The process is in statistical equilibrium.

Why does Stationarity Assumption work?

- Now, suppose each distribution has same mean. In that case the common mean could be estimated based on the realization of size ‘n’.
- We can visualize the fact in the following way---

Suppose we have 10 identical machines producing some item, *say*, bulb. Suppose each machine is run for one hour. Now it is easy to visualize that total (average) output by 10 machines is same as that of total (average) output by a single machine running for 10 hours.

TYPES OF STATIONARITY

- **STRICT (STRONG OR COMPLETE) STATIONARY PROCESS:** Consider a finite set of r.v.s. $Y_{t_1}, Y_{t_2}, \ldots, Y_{t_n}$ from a stochastic process $\{Y(w, t); t = 0, \pm 1, \pm 2, \ldots\}$.

- The n-dimensional distribution function is defined by $F_{F_{t_1}, F_{t_2}, \ldots, F_{t_n}}(y_{t_1}, y_{t_2}, \ldots, y_{t_n}) = P(w: Y_{t_1} < y_1, \ldots, Y_{t_1} < y_n)$

where y_i, $i = 1, 2, \ldots, n$ are any real numbers.

STRONG STATIONARITY

- A process is said to be **first order stationary** in distribution, if its one dimensional distribution function is time-invariant,

  i.e., $F_{F_{t_1}}(y_1) = F_{F_{t_{1+k}}}(y_1)$ for any t_1 and k.

- **Second order stationary** in distribution if $F_{F_{t_1},F_{t_2}}(y_1, y_2) =$ $F_{F_{t_{1+k}},F_{t_{2+k}}}(y_1, y_2)$ for any t_1, t_2 and k.

- **$\mathbf{n^{th}}$ order stationary** in distribution if

  $F_{F_{t_1},F_{t_2},\ldots,F_{t_n}}(y_1, y_2, \ldots, y_n) = F_{F_{t_{1+k}},F_{t_{2+k}},\ldots,F_{t_{n+k}}}(y_1, y_2, \ldots, y_n)$

  for any $t_1, \ldots, t_n$ and k.

STRONG STATIONARITY

n^{th} order stationarity in distribution = strong stationarity

Shifting the time origin by an amount "k" has no effect on the joint distribution, which must therefore depend only on time intervals between $t_1, t_2 \ldots, t_n$ not on absolute time, t.

WEAK STATIONARITY

- WEAK (COVARIANCE) STATIONARITY OR STATIONARITY IN WIDE SENSE: A time series is said to be **covariance stationary** if its first and second order moments are unaffected by a change of time origin.
- That is, we have constant mean and variance with covariance and correlation beings functions of the time difference only.

WEAK STATIONARITY

$$E[y_t] = \mu, \qquad \text{✯}t$$

$$var[y_t] = \sigma^2 < \infty, \qquad \text{✯}t$$

$$cov[y_t, y_{t-k}] = \gamma_k, \qquad \text{✯}t$$

$$corr[y_t, y_{t-k}] = \rho_k, \qquad \text{✯}t$$

From, now on, when we say “stationary”, we imply weak stationarity.

Example of First-Order Stationarity

RANDOM PROCESS is $x(t) = A\sin(\omega_0 t + \theta_0)$

- Assume that A and ω_0 are constants; θ_0 is a uniformly distributed RV from $[-\pi,\pi)$; t is time.
- From last lecture, recall that the PDF of $x(t)$:

$$f_x(x) = \begin{cases} \dfrac{1}{\pi\sqrt{A^2 - x^2}} & |x| \leq A \\ 0 & x \text{ Elsewhere} \end{cases}$$

- *Note*: there is NO dependence on time, the PDF is not a function of t.
- The RP is STATIONARY.

Non-Stationary Example

RANDOM PROCESS is $x(t) = A\sin(\omega_0 t + \theta_0)$

- Now assume that A, θ_0 and ω_0 are constants; t is time.
- Value of x(t) is <u>always</u> known for any time with a probability of 1. Thus the first order PDF of x(t) is

$$f(x) = \delta(x - A\sin(\omega_0 t + \theta_0))$$

- *Note*: The PDF depends on time, so it is NONSTATIONARY.

Ergodic Processes

- Definition: A random process is *ERGODIC* if all time averages of any sample function are equal to the corresponding ensemble averages (expectations)
- Example, for ergodic processes, can use ensemble statistics to compute DC values and RMS values

$$x_{DC} = \langle x(t) \rangle = \overline{[x(t)]} = m_x$$

$$\langle x(t) \rangle = \lim_{T \to \infty} \frac{1}{T} \int_0^T [x(t)]dt \qquad \text{Time average}$$

$$\overline{[x(t)]} = \int_{-\infty}^{\infty} [x] f(x) dx = m_x \qquad \text{Ensmble average}$$

$$x_{RMS} = \sqrt{\langle x^2(t) \rangle} = \sqrt{\overline{x^2}} = \sqrt{\sigma^2 + m_x^2}$$

- Ergodic processes are always stationary; Stationary processes are not necessarily ergodic

$$\boxed{\text{Ergodic} \Rightarrow \text{Stationary}}$$

Example: Ergodic Process

RANDOM PROCESS is $x(t) = A\sin(\omega_0 t + \theta_0)$

- A and ω_0 are constants; θ_0 is a uniformly distributed RV from $[-\pi,\pi)$; t is time.
- Mean (Ensemble statistics)

$$m_x = \overline{x} = \int_{\infty}^{\infty} x(\theta) f_\theta(\theta) d\theta = \int_{\pi}^{\pi} A\sin(\omega_0 t + \theta)\frac{1}{2\pi} d\theta = 0$$

- Variance

$$\sigma_x^{\ 2} = \int_{\pi}^{\pi} A^2 \sin^2(\omega_0 t + \theta)\frac{1}{2\pi} d\theta = \frac{A^2}{2}$$

Example: Ergodic Process

- Mean (Time Average) T is large

$$\langle x(t)\rangle = \lim_{T\to\infty}\frac{1}{T}\int_0^T A\sin(\omega_0 t+\theta)\,dt = 0$$

- Variance

$$\langle x^2(t)\rangle = \lim_{T\to\infty}\frac{1}{T}\int_0^T A^2\sin^2(\omega_0 t+\theta)\,dt = \frac{A^2}{2}$$

- The ensemble and time averages are the same, so the process is ERGODIC

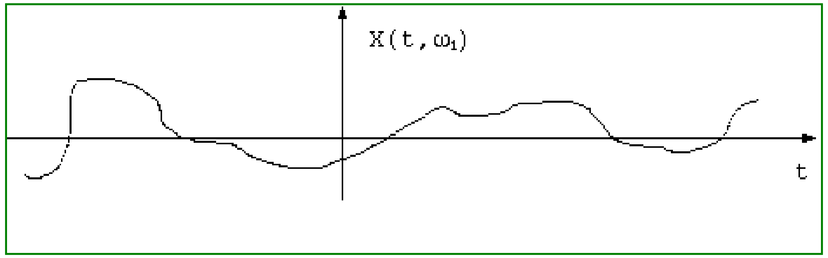
EXERCISE

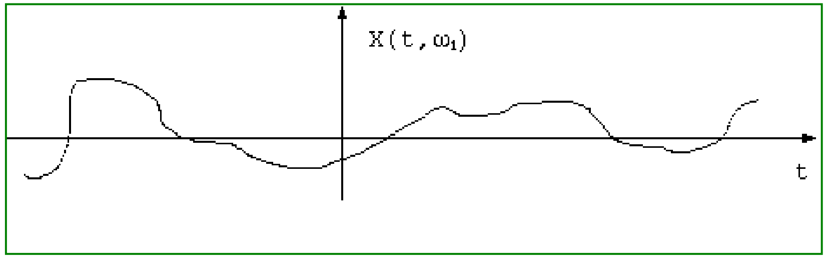
- Write down the definition of :
  - Wide sense stationary
  - Ergodic processes
- How do these concepts relate to each other?
- Consider: $x(t) = K$; K is uniformly distributed between [-1, 1]
  - WSS?
  - Ergodic?

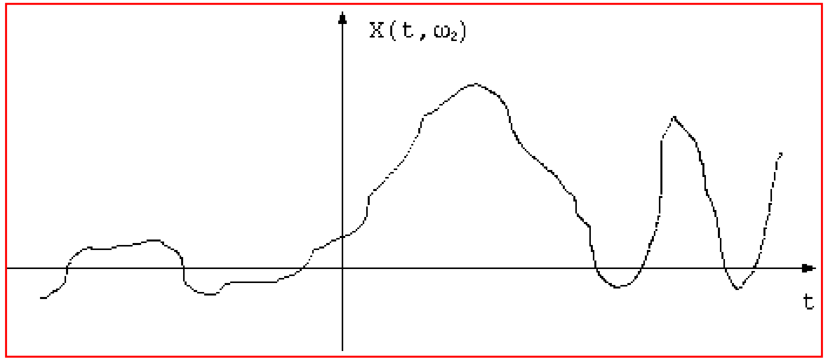
Autocorrelation of Random Process

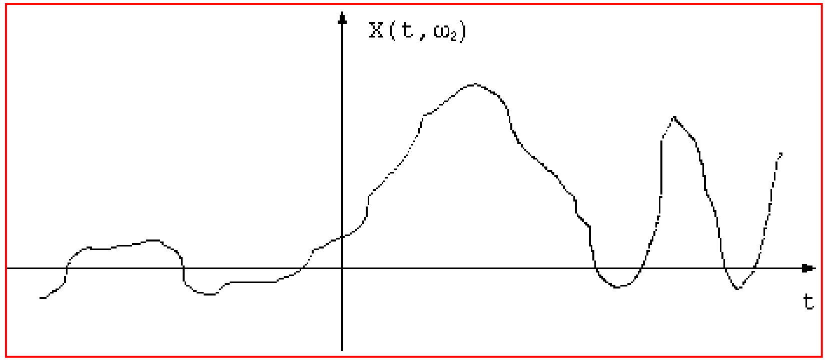
- The Autocorrelation function of a real random process x(t) at two times is:

$$R_x(t_1,t_2) = \overline{x(t_1)x(t_2)} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 f_x(x_1,x_2)\,dx_1 dx_2$$









Wide-sense Stationary

- A random process that is stationary to order 2 or greater is *Wide-Sense Stationary*:
- A random process is *Wide-Sense Stationary* if:

$$\overline{x(t)} = \text{constant}$$

$$R_x(t_1, t_2) = R_x(\tau)$$

- Usually, $t_1 = t$ and $t_2 = t + \tau$ so that $t_2 - t_1 = \tau$.
- Wide-sense stationary process does not DRIFT with time.
- Autocorrelation depends only on the time gap but not where the time difference is.
- Autocorrelation gives idea about the frequency response of the RP.

Autocorrelation Function of RP

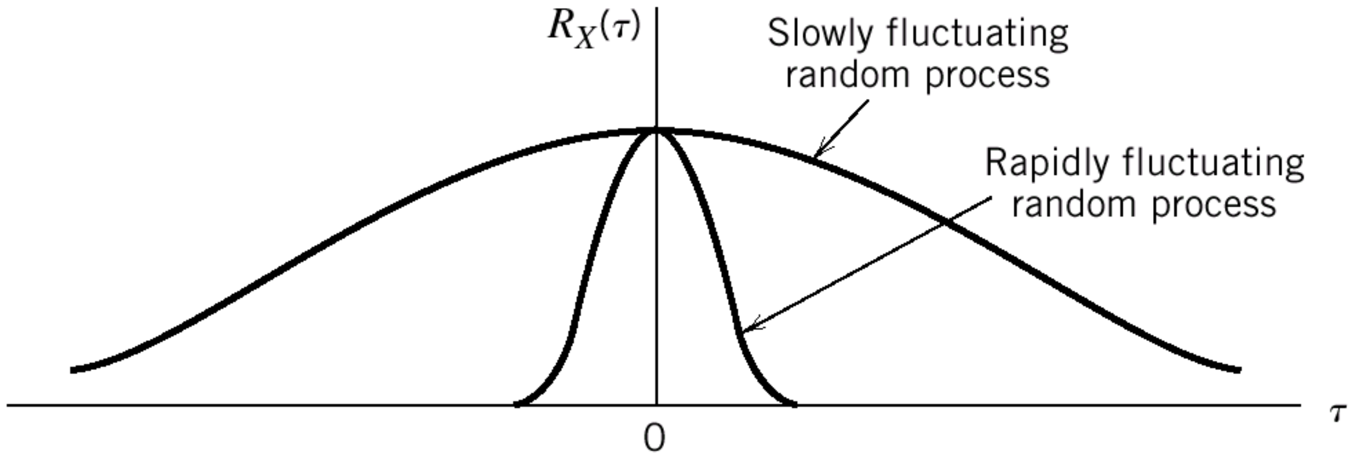
- Properties of the autocorrelation function of wide-sense stationary processes

$$R_x(0) = \overline{x^2(t)} = \text{Second Moment}$$

$$R_x(\tau) = R_x(-\tau), \quad \text{Symmetric}$$

$$R_x(0) \geq \left|R_x(\tau)\right|, \quad \text{Maximum value at 0}$$

Autocorrelation of slowly and rapidly fluctuating random processes.



Properties of stationary processes

Mean, Variance and Wide-sense Stationarity

Mean

The mean of a stationary process and its estimate are defined as:

$$\boxed{\mu_x = E(x_t) = \int_{-\infty}^{\infty} x f(x)\, dx \qquad \bar{x} = \frac{1}{N}\sum_{k=1}^{N} x(t_k)}$$

Variance

The variance of a stationary process and its estimate are defined as:

$$\boxed{\sigma_x^2 = E((x_t - \mu_x)^2) = \int_{-\infty}^{\infty} (x_t - \mu_x)^2 f(x)\, dx \qquad \hat{\sigma}_x^2 = \frac{1}{N-1}\sum_{k=1}^{N} (x(t_k) - \bar{x})^2}$$

Auto-Covariance function (ACF)

- The ACF of a time-series (i.e., inter-dependence of samples of time-series) is defined as $\sigma(t,s) = E((x_t - \mu_t)(x_s - \mu_s))$
- For stationary time-series, the mean is constant and the dependence is only a function of the lag $l = t - s$.
- The ACF of a stationary process and its estimate are given by:

$$
\begin{aligned}
\sigma_{xx}[l] &= E((x[t]-\mu_x)(x[t+l]-\mu_x)) \\
\hat{\sigma}_{xx}[l] = r_{xx}[l] &= \frac{1}{N}\sum_{k=1}^{N-l}(x[k]-\bar{x})(x[k+l]-\bar{x})
\end{aligned}
$$

Properties of ACF

- ACF of a time-series is symmetric about the lag l .
- It takes the largest value at lag $l = 0$.
- In practice, a normalized function known as the auto-correlation function is used:
  - $\rho_{xx}[l] = \frac{\sigma_{xx}[l]}{\sigma_{xx}[0]}$

Cross Correlations of RP

- Cross Correlation of two RP $x(t)$ and $y(t)$ is defined similarly as:

$$R_{xy}(t_1,t_2) = \overline{x(t_1)y(t_2)} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 y_2 f_{xy}(x_1,y_2)\,dx_1 dy_2$$

- If $x(t)$ and $y(t)$ are Jointly Stationary processes,

$$R_{xy}(t_1,t_2) = R_{xy}(t_2 - t_1) = R_{xy}(\tau) \qquad \tau = t_2 - t_1$$

- If the RP's are jointly ERGODIC,

$$R_{xy}(\tau) = \overline{x(t)y(t+\tau)} = \langle x(t)y(t+\tau)\rangle$$

Cross Correlation Properties of Jointly Stationary RP's

- Some properties of cross-correlation functions are

$$R_{xy}(\tau) = R_{xy}(-\tau)$$

$$\left|R_{xy}(\tau)\right| \leq \sqrt{R_x(0) R_y(0)}$$

$$\left|R_{xy}(\tau)\right| \leq \tfrac{1}{2}\left[R_x(0) + R_y(0)\right]$$

- Uncorrelated:

$$R_{\mathrm{xy}}(\tau) = \overline{x(t)\, y(t+\tau)} = \overline{x}\;\overline{y}$$

- Orthogonal:

$$R_{\mathrm{xy}}(\tau) = 0$$

- Independent: if $x(t_1)$ and $y(t_2)$ are independent (joint distribution is product of individual distributions)

Cross-Covariance function (CCF)

- The CCF is a measure of dependence between samples of a time-series $\{x(k)\}$ and samples of another time-series $\{y(s)\}$. For stationary series, this is only a function of the distance between samples, i.e., the lag l .
- The CCF of a stationary process and its estimate are given by:

$$
\begin{aligned}
\sigma_{xy}[l] &= E((x[t]-\mu_x)(y[t+l]-\mu_y)) \\
\hat{\sigma}_{xy}[l] = r_{xy}[l] &= \frac{1}{N}\sum_{k=1}^{N-l}(x[k]-\bar{x})(y[k+l]-\bar{y})
\end{aligned}
$$

Properties of CCF

- CCF of two time-series satisfy $\sigma_{xx}[l] = \sigma_{xx}[-l]$
- The largest value occurs at lag where the dependency is strongest.
- In practice, a normalized function known as the cross-correlation function is used.
  - $\rho_{xy}[l] = \frac{\sigma_{xx}[l]}{\sqrt{\sigma_{xx}[0]\sigma_{yy}[0]}}$

Uses of ACF and CCF

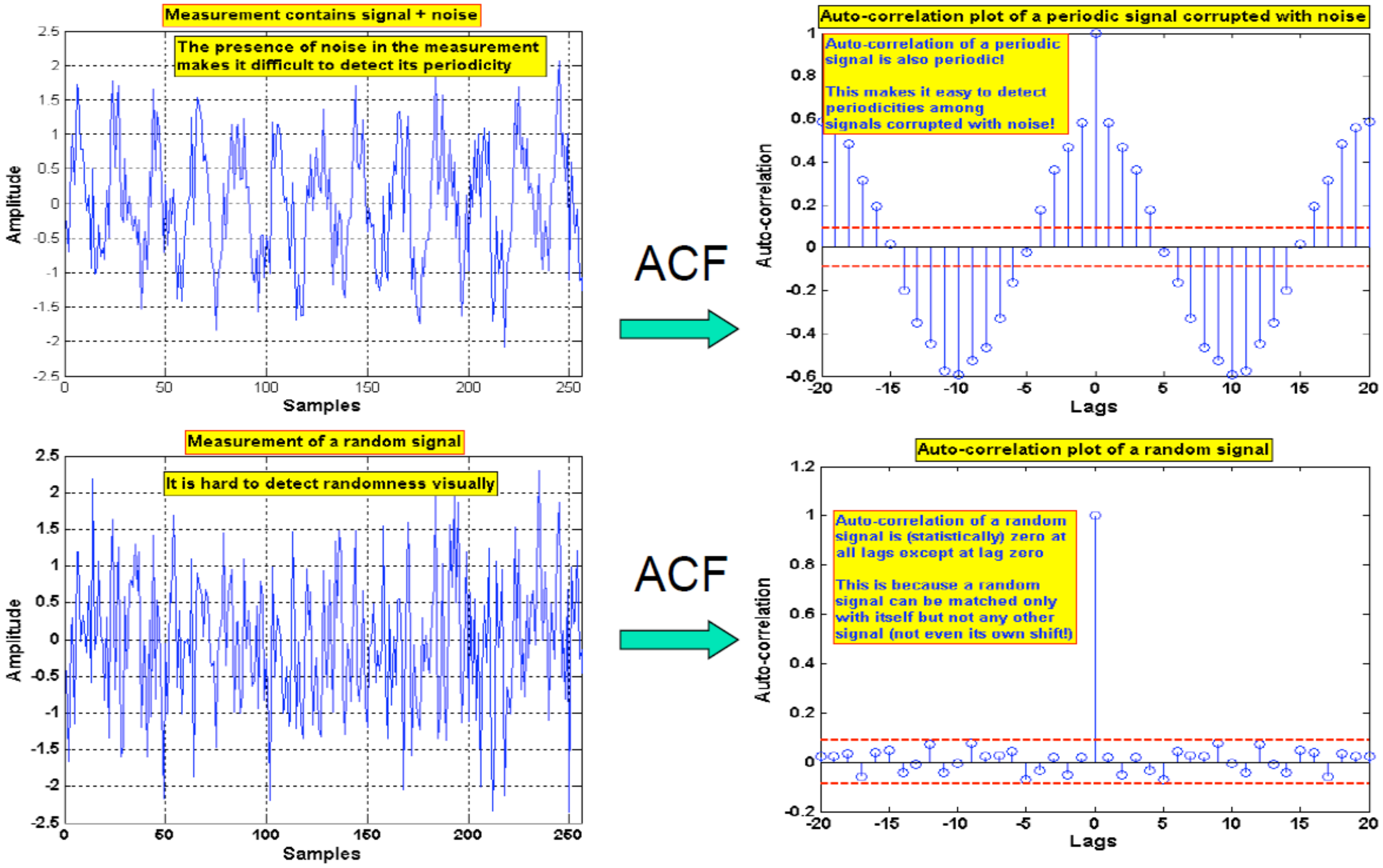
ACF and CCF are also defined for deterministic sequences in a similar way.

They have wide uses in system identification:

Applications

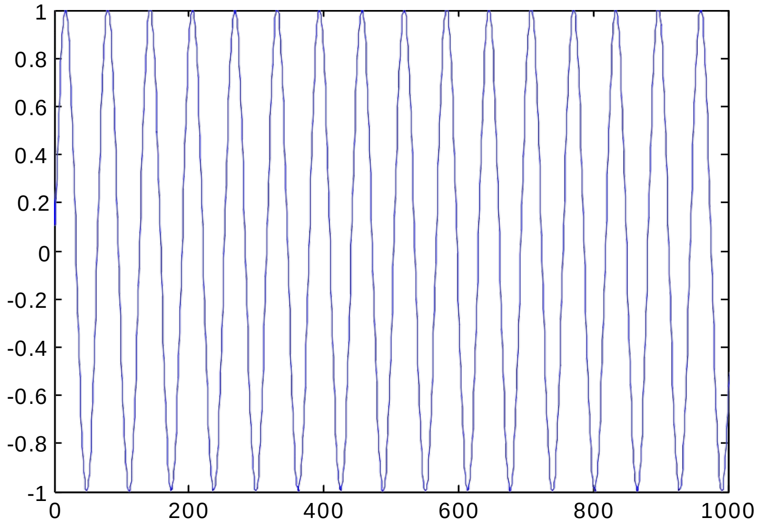
- The ACF is used in detecting the underlying process, *i.e.*, whether it is periodic, moving average, auto-regressive, independent, etc.
- ACF is used in the estimation of periodicity of an oscillatory process and order of certain stochastic models.
- The CCF assumes the largest value when two time-series have strongest correlation. This has wide application in determining the time-delay of a system (important step in identification)
- The cross-correlation function is the output of an LTI system whose input is the auto-correlation function. This relationship is used in the estimation of impulse response coefficients.

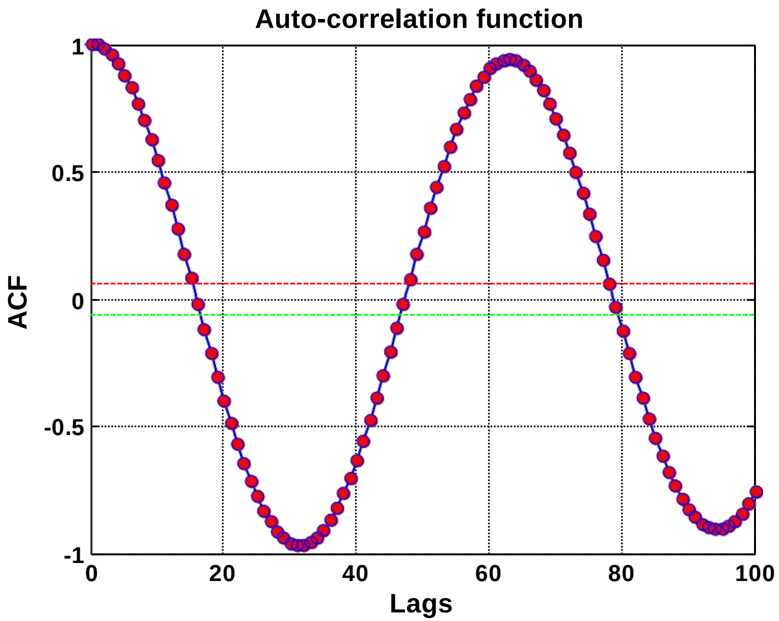
Use of auto-correlation function: examples



Auto-correlation Function

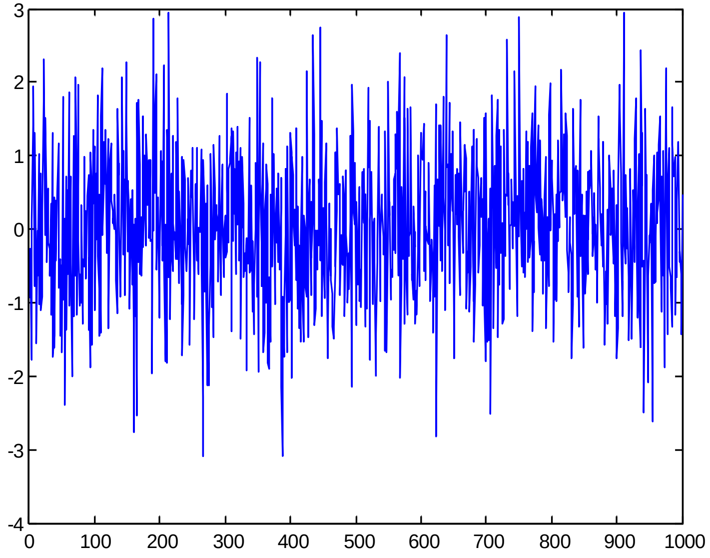
- ACF of a sinusoid with itself

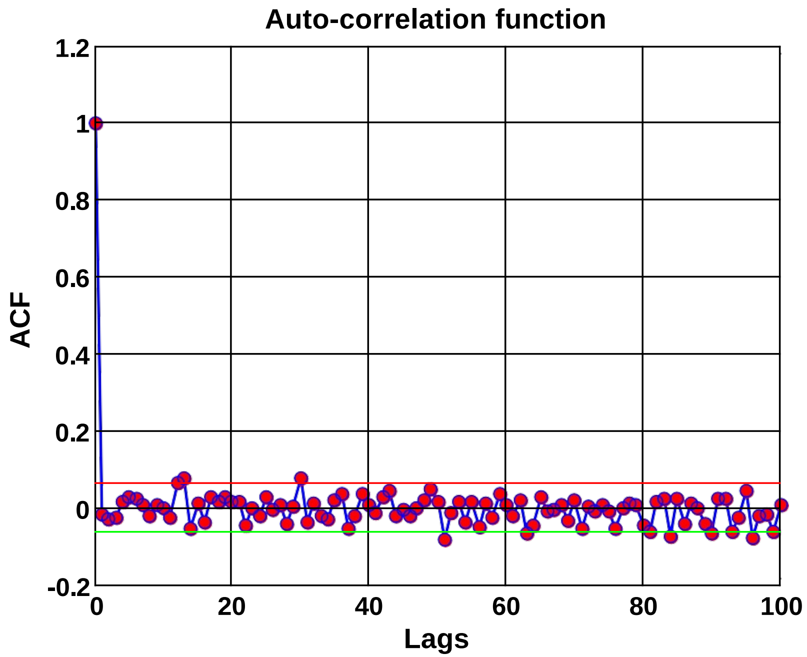




Autocorrelation Function

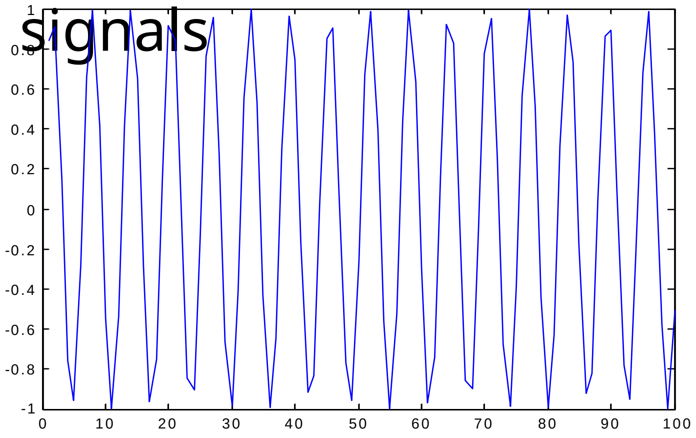
- ACF of a random data set – zero correlation with itself – indicative of absence of predictability (white data set)

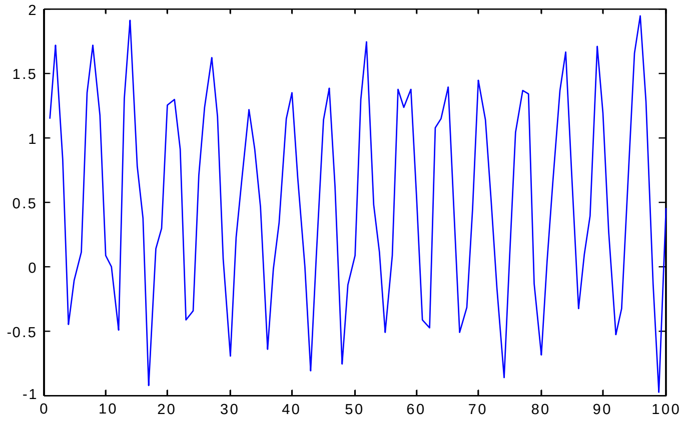


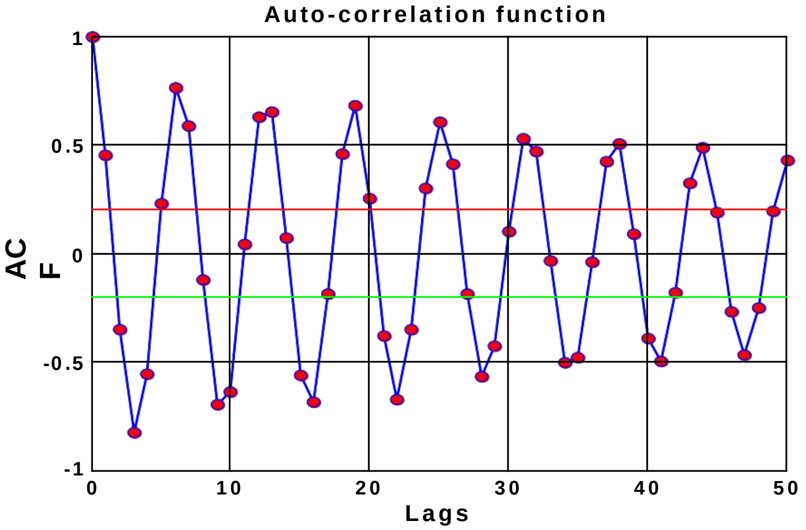


An application of ACF

- ACF can be used to detect oscillations in noisy signals







Oscillations of the same frequency